

# Unsupervised Learning Project: A Reproducible Experimental Study of Clustering

## Milestone 2

### Front matter

|                                 |                                                                                                                                                   |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Team                            | G4/P1 - José Moutinho-73129, Afonso Simões-73204, Luís Nunes-73216                                                                                |
| Project title                   | Booking Patterns for Cancellation Risk Management                                                                                                 |
| Checkpoint date                 | 30 April 2026                                                                                                                                     |
| Repository                      | <a href="https://github.com/AfonsoCSimoes/UnsupervisedLearning-Project.git">https://github.com/AfonsoCSimoes/UnsupervisedLearning-Project.git</a> |
| Repository version shown in lab | Commit hash: fea6280<br>Branch name: main                                                                                                         |

### 1. RQ1 and segmentation time

- Concrete segmentation question: "Do bookings made far in advance through indirect channels naturally emerge as a distinct pre-arrival cluster, and how does this profile associate with observed cancellation rates to inform risk-mitigation policies?"
- Unit of analysis: One row = one individual hotel booking record.
- Segmentation time (index time): Exactly immediately after the initial booking is recorded. Clustering features are limited to the snapshot available at this precise timestamp.

### 2. Dataset version control and documentation

- Course dataset release/version used: "course release v1" (Hotel Booking Demand).
- Dataset file fingerprint: SHA-256 hash: 7c2ae42a7353905ea136e5c2287f17c92c5435826598bfbb8491c6f0c7b1fc06.
- Data-quality issues identified:
  - Minor missingness in the categorical country variable (approx. 0.4%).
  - Data entry errors: Negative Average Daily Rate (ADR = -6.38).
  - Extreme heavy-tails/outliers: ADR maxing at 5400.00, requiring filtering to prevent Euclidean space distortion.
  - "NULL" strings in agent and company fields representing "not applicable" rather than standard missing data.

### 3. Completed Exploratory Data Analysis (EDA)

- Numerical missingness summary: Missing data is negligible. Categorical *country* contains 488 missing values (0.4%). Numerical *children* contains 4 missing values (0.003%).
- Outlier summary: Core numerical variables (*lead\_time*, *previous\_cancellations*) show high right-skewness and heavy tails, needing logarithmic transformation prior to scaling. High monetary outliers (*ADR* > 5000) and negative values were filtered.

- Variables dropped before modelling: *agent* and *company* were excluded due to extreme cardinality as raw identifiers. *arrival\_date\_year* was dropped as a chronological artifact to prevent artificial separation of identical inter-year booking behaviors.
- Leakage exclusions: *is\_canceled*, *reservation\_status*, and *reservation\_status\_date* were excluded as outcome variables. Operational variables generated post-booking (*assigned\_room\_type*, *booking\_changes*) were excluded to enforce the index-time constraint. These are reserved for post-hoc profiling.

#### 4. Main representation (RQ2)

- Included variables: *lead\_time*, *total\_nights* (derived from weekend/week nights), *party\_size* (derived from adults/children/babies), *distribution\_channel*, *market\_segment*, *customer\_type*, *deposit\_type*, *hotel*, *arrival\_date\_month*, and *country\_grouped*.
- Excluded/profiling-only variables: *adr* (Average Daily Rate) and the leakage exclusions listed in Section 3 are extracted pre-pipeline. *adr* is excluded from the input matrix to ensure clustering measures behavioral similarity rather than price.
- Numerical preprocessing: Missing *children* values imputed with 0. Right-skewed variables (*lead\_time*, *previous\_cancellations*, *previous\_bookings\_not\_canceled*) processed via log1p transformation. All numerical variables are variance-standardized using *StandardScaler* to equalize distance contributions.
- Categorical preprocessing: The high-cardinality *country* variable was grouped into a "Top 15 + Other" policy. Nominal variables (including seasonality via month) were processed using full One-Hot Encoding. Dummy columns remain as unscaled (0/1) coordinates to prevent rare categories to distort the Euclidean distance.
- Final representation\_id: R-EUCLID-standard-countryTop15-noADR.

#### 5. Baseline modelling evidence

- Methods: K-means and iK-means (Anomalous Pattern extraction).
- Fixed K values: A bounded grid of  $K \in [3, 8]$  was evaluated. This range is plausible for segmenting hotel demand into distinct behavioral profiles without over-fragmenting the data into noisy, micro-clusters.
- Internal indices: Computed in the R-EUCLID space, K-means at  $K=5$  yielded the optimal coherent partition: Silhouette Score (0.1544), Davies-Bouldin (1.8157), and Calinski-Harabasz (15647.03). iK-means successfully mathematically derived 14 natural anomalous patterns, validating the existence of distinct sub-groups.
- Runtime: K-means executed efficiently on the full dataset across multiple seeds. The deterministic iK-means required slightly higher initialization compute to sequentially extract maximum-distance anomalous patterns. Full runtime: 13 minutes, 11 seconds.
- First cluster-interpretation note: A post-hoc analysis ( $\Delta = 100 \times \frac{\text{cluster\_mean} - \text{grand\_mean}}{\text{grand\_mean}}$ ) applied to the unscaled features validates the  $K=5$  topology. Cluster 3 perfectly isolates the hypothesized profile: extreme early planners (*lead\_time* = +104.6%) using indirect channels (*is\_Direct* = -79.2%). Profiling reveals this segment exhibits a cancellation rate 167.0% higher than the dataset average, directly answering RQ1 and confirming the need for targeted risk policies.

#### 6. Sensitivity / robustness plan already defined

- Main representation: R-EUCLID-standard-countryTop15-noADR.
- Resampling rule: To validate topological stability, a bootstrap resampling procedure was implemented with 20 independent bootstrap runs ( $n_{\text{boot}}=20$ ) for the K grid.
- Planned stability measure: Mean pairwise Adjusted Rand Index (ARI) computed across the resampled cluster assignments.

- Question answered by comparison: Are the discovered clusters genuine behavioral variations, or artifacts of extreme numerical values fluctuations? The optimal K=5 baseline achieved a highly stable mean ARI of 0.921 and standard deviation of 0.174. This stability held identically when tested against a RobustScaler variant, confirming the heavy tails do not dominate the distance space and the standard scaling is methodologically robust.

## 7. Experiment logging and repository status

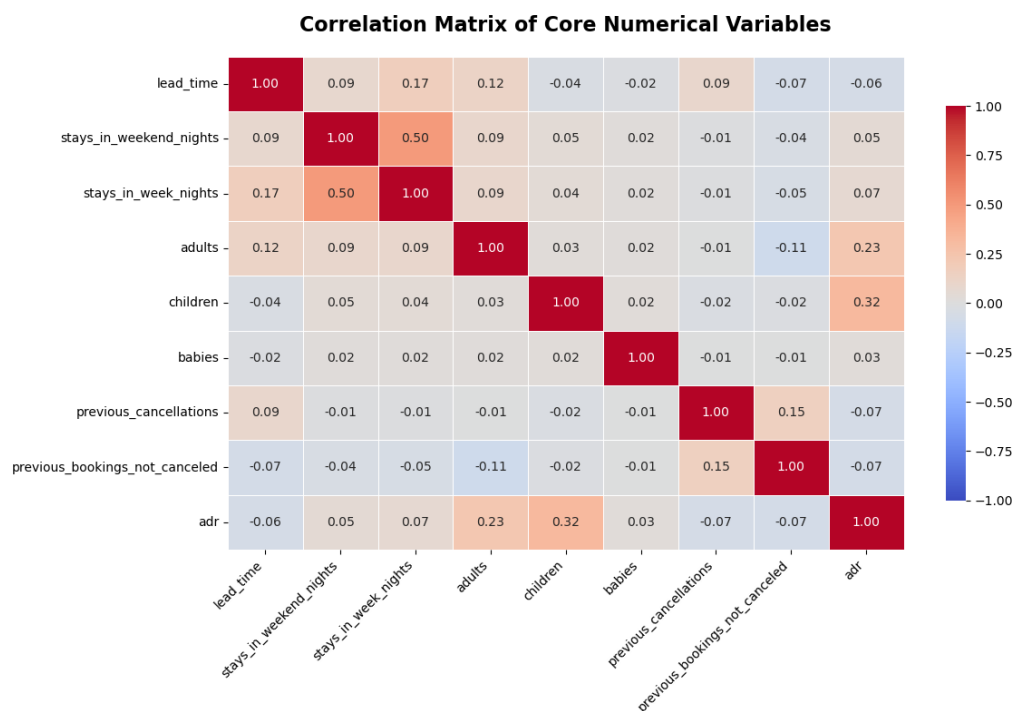
- Experiment schema: The pipeline results are logged in the experiments.csv file. Columns: representation\_id, method, k, seed, silhouette, calinski\_harabasz, and davies\_bouldin. Stability indices (mean\_ARI, std\_ARI) are logged in the notebook file dedicated to the baseline run.
- Environment: An environment.yml file exists to ensure dependency parity.
- Run command: A single-entry point script (run\_all.py) is fully functional. It executes data loading, extraction of post-hoc profiling variables, preprocessing (generating the R-EUCLID matrix), and the complete model evaluation loop, appending results directly to the CSV logs.

## Appendix

- Outlier and Heavy Tail Summary:

|       | lead_time | stays_in_weekend_nights | stays_in_week_nights | adults    | children | babies    | previous_cancellations | previous_bookings_not_canceled | adr       |
|-------|-----------|-------------------------|----------------------|-----------|----------|-----------|------------------------|--------------------------------|-----------|
| count | 119390.00 | 119390.00               | 119390.00            | 119390.00 | 119386.0 | 119390.00 | 119390.00              | 119390.00                      | 119390.00 |
| mean  | 104.01    | 0.93                    | 2.50                 | 1.86      | 0.1      | 0.01      | 0.09                   | 0.14                           | 101.83    |
| std   | 106.86    | 1.00                    | 1.91                 | 0.58      | 0.4      | 0.10      | 0.84                   | 1.50                           | 50.54     |
| min   | 0.00      | 0.00                    | 0.00                 | 0.00      | 0.0      | 0.00      | 0.00                   | 0.00                           | -6.38     |
| 25%   | 18.00     | 0.00                    | 1.00                 | 2.00      | 0.0      | 0.00      | 0.00                   | 0.00                           | 69.29     |
| 50%   | 69.00     | 1.00                    | 2.00                 | 2.00      | 0.0      | 0.00      | 0.00                   | 0.00                           | 94.58     |
| 75%   | 160.00    | 2.00                    | 3.00                 | 2.00      | 0.0      | 0.00      | 0.00                   | 0.00                           | 126.00    |
| max   | 737.00    | 19.00                   | 50.00                | 55.00     | 10.0     | 10.00     | 26.00                  | 72.00                          | 5400.00   |

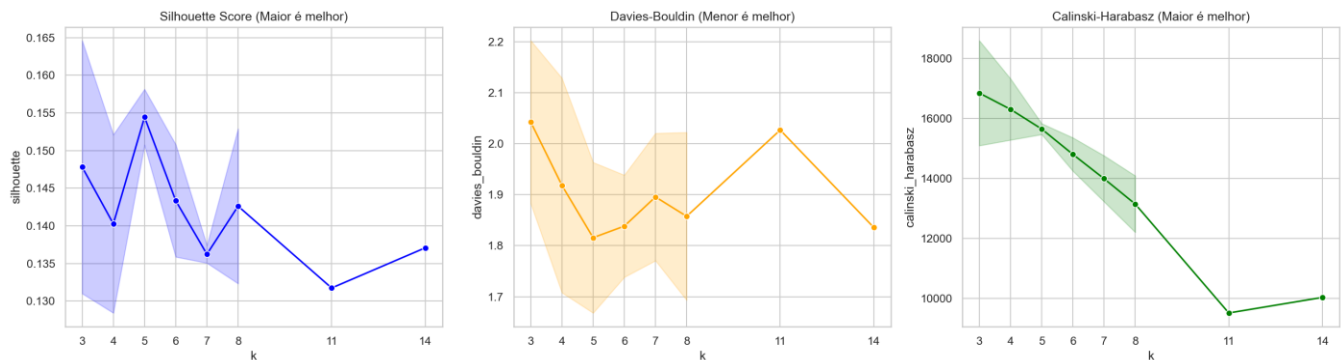
- Correlation Matrix of Core Numerical Variables:



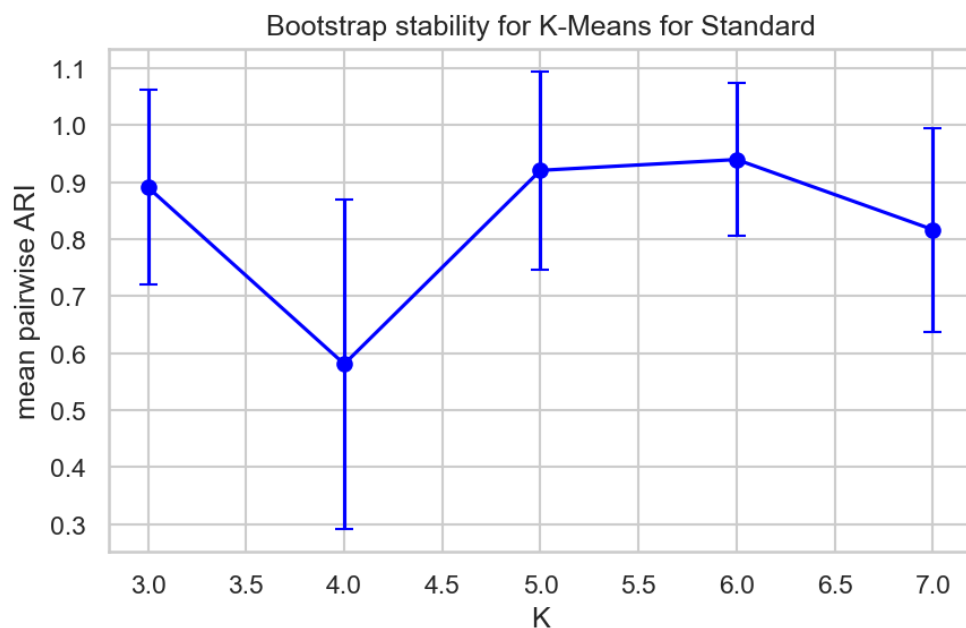
• Feature-selection table:

| Variable                                      | Action    | Rationale                                                                             |
|-----------------------------------------------|-----------|---------------------------------------------------------------------------------------|
| lead_time                                     | Include   | Key behavioral indicator; log-transformed to handle heavy tail.                       |
| stays_in_weekend_nights, stays_in_week_nights | Derived   | Combined into total_nights to capture stay duration.                                  |
| adults, children, babies                      | Derived   | Combined into party_size to represent total guests.                                   |
| distribution_channel, market_segment          | Include   | Essential for RQ1; One-Hot Encoded (OHE).                                             |
| country                                       | Include   | Grouped into Top 15 + 'Other' to manage cardinality.                                  |
| is_canceled, reservation_status               | Exclude   | <b>Leakage Control:</b> Outcome variables realized after booking.                     |
| reservation_status_date                       | Exclude   | Post-event attribute; realized after arrival/cancellation.                            |
| agent, company                                | Exclude   | High-cardinality identifiers; dropped to prevent overfit.                             |
| adr                                           | Profiling | Excluded from clustering inputs to avoid price-dominance; used for post-hoc analysis. |

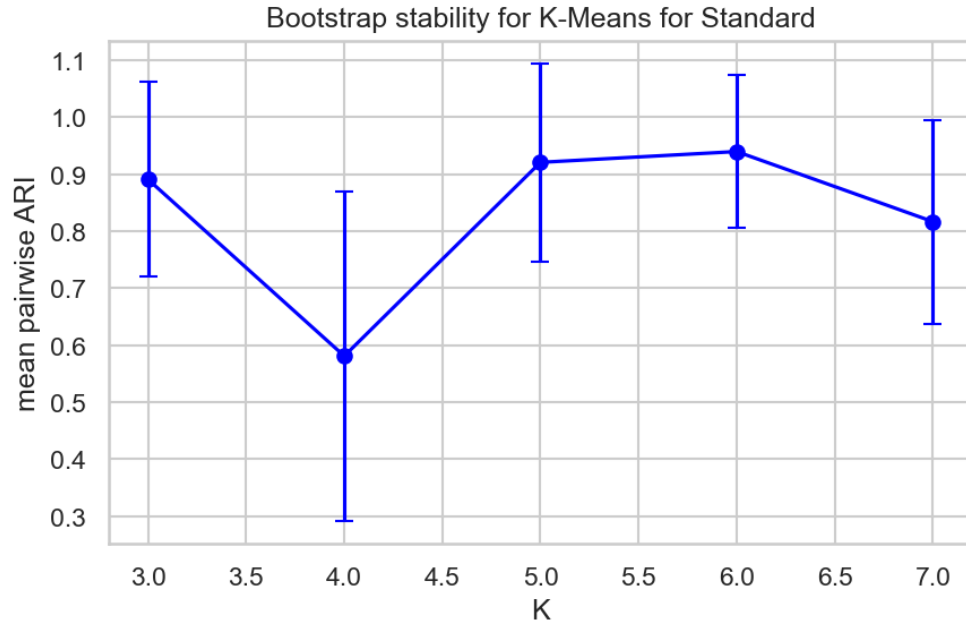
• Feature-selection table:



• Bootstrap Stability for K-Means for Standard



• Bootstrap Stability for K-Means for Robust:



• Cluster Interpretation: Relative Differences:

| Cluster                | 0      | 1      | 2      | 3       | 4       |
|------------------------|--------|--------|--------|---------|---------|
| lead_time              | -85.3% | -92.8% | 28.4%  | 104.6%  | 24.2%   |
| total_nights           | -49.8% | -41.7% | 126.6% | -20.1%  | -14.5%  |
| party_size             | -38.0% | -12.5% | 13.7%  | -5.9%   | 3.1%    |
| previous_cancellations | 567.3% | -99.9% | -97.2% | 1687.9% | -100.0% |
| is_TA_TO               | -81.7% | -28.3% | 1.5%   | 12.4%   | 12.9%   |
| is_Direct              | 74.2%  | 128.2% | 19.1%  | -79.2%  | -54.2%  |
| is_Transient           | 19.7%  | 8.2%   | 3.5%   | -31.1%  | -2.3%   |
| is_Non_Refund          | -96.4% | -92.4% | -86.3% | 404.5%  | 29.5%   |
| adr                    | -39.4% | -7.7%  | 10.8%  | -23.6%  | 3.6%    |
| is_canceled            | -85.0% | -62.2% | -8.8%  | 167.0%  | 16.2%   |

## • experiments.csv excerpt:

```
representation_id,method,k,seed,silhouette,calinski_harabasz,davies_bouldin,runtime
R-EUCLID-standard-countryTop15-noADR,k-means,3,42,0.16736430088384502,15308.416026377978,1.9058052390461553,4.2999560832977295
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